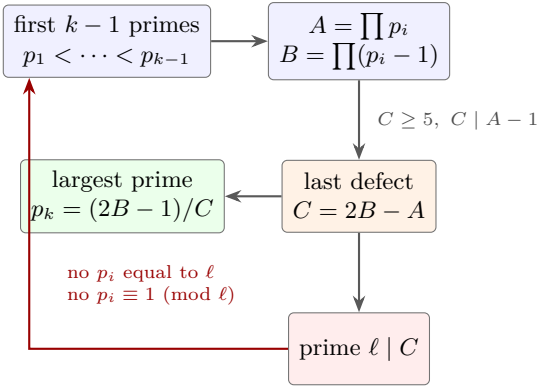
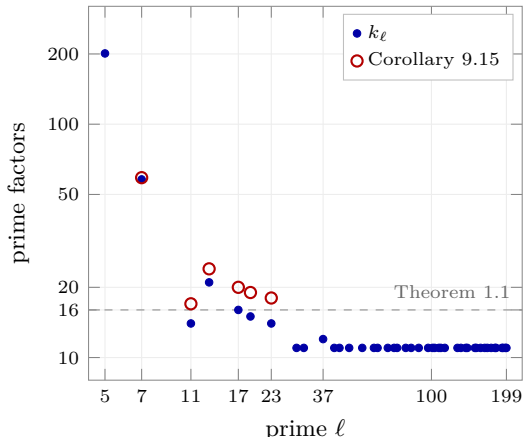
(a)


first $k-1$ primes
$p_1 < \cdots < p_{k-1}$
$A = \prod p_i$
$B = \prod (p_i - 1)$
$C \geq 5,\ C \mid A - 1$
last defect
$C = 2B - A$
largest prime
$p_k = (2B-1)/C$
no p_i equal to ℓ
no $p_i \equiv 1 \pmod{\ell}$
prime $\ell \mid C$

(b)


k_ℓ
Corollary 9.15
Theorem 1.1
prime factors
200
100
50
20
16
10
5
7
11
17
23
37
100
199
prime ℓ